



C20-M-505

7659

BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DME – FIFTH SEMESTER EXAMINATION

REFRIGERATION AND AIR CONDITIONING

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define the terms (a) refrigeration, (b) ton of refrigeration and (c) coefficient of performance. 3×1
2. A reversed Carnot cycle refrigeration system works between the temperature limits of 37 °C and –3 °C. Calculate COP of the system. 3
3. State the purpose of flash chamber and accumulator in the vapour compression system. 1½+1½
4. State any three differences between primary and secondary refrigerants. 3×1
5. What are the functions of (a) expansion device and (b) compressor in refrigerating system? 1½+1½
6. What is the function of drier in refrigeration system? List the types of driers. 1+2
7. Define the terms (a) relative humidity, (b) dew point temperature and (c) dry bulb temperature. 3×1
8. List any six of psychrometric processes. 6×½
9. State any three advantages of forced draft cooling tower over natural draft cooling tower? 3×1
10. State any three advantages of unitary air conditioning system. 3×1

PART—B

8×5=40

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **eight** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11. (a)** A refrigeration system produces 40 kg/hr of ice at 0 °C from water at 25 °C. Find the refrigeration effect per hour and TR. If it consumes 1 kW of energy to produce the ice, find the COP. Take latent heat of solidification of water at 0 °C as 335 kJ/kg and specific heat of water 4.19 kJ/kg °C. 8

(OR)

- (b)** Explain Bell Coleman air refrigeration cycle with neat sketch and draw P-V and T-S diagrams. 6+2

- 12. (a)** Explain the effect of sub cooling and super heating of refrigerant on COP of vapour compression refrigeration system with the help of P-H diagram. 4+4

(OR)

- (b)** Explain the working of solar powered vapour absorption refrigeration system with neat sketch. 4+4

- 13. (a)** Explain the working principle of flooded-type evaporator with a neat sketch. 4+4

(OR)

- (b)** Explain the working of cold storage plant with neat sketch. 4+4

- 14. (a)** Explain sling psychrometer with a neat sketch. 4+4

(OR)

- (b)** Find dew point temperature, relative humidity and specific humidity of moist air at dry bulb temperature of 30 °C and wet bulb temperature of 20 °C. 8

- 15. (a)** Explain the working of winter air conditioning system with the help of neat sketch. 4+4

(OR)

- (b) Explain the working of year-round air-conditioning system with a neat sketch.

4+4

PART—C

10×1=10

- Instructions :** (1) Answer the following question.
(2) The question carries **ten** marks.
(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

- 16.** A vapour compression refrigerator uses methyl chloride (R-40) and operates between temperature limits of -10°C and 45°C . At entry to the compressor, the refrigerant is dry saturated and after compression it acquires a temperature of 60°C . Find the C.O.P. of the refrigerator. The relevant properties of methyl chloride are as follows :

Saturation Temperatrue $^{\circ}\text{C}$	Enthalpy in kJ/kg		Entropy in kJ/kgK	
	Vapour	Liquid	Vapour	Liquid
-10	460.7	45.4	0.183	1.637
45	483.6	133.0	0.485	1.587

Take C_p for R-40 as 0.615 kJ/kgK .

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